

Proteomic and functional comparison between human induced and embryonic stem cells

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Abstract

Human induced pluripotent stem cells (hiPSCs) have great potential to be used as alternatives to embryonic stem cells (hESCs) in regenerative medicine and disease modelling, thereby avoiding many of the ethical issues arising from the use of embryo-derived cells. However, despite clear similarities between the two cell types, it is likely they are not identical. In this study, we characterise the proteomes of multiple hiPSC and hESC lines derived from independent donors. We find that while hESCs and hiPSCs express a near identical set of proteins, they show consistent quantitative differences in the expression levels of a wide subset of proteins.

hiPSCs have increased total protein content, while maintaining a comparable cell cycle profile to hESCs. The proteomic data show hiPSCs have significantly increased abundance of vital cytoplasmic and mitochondrial proteins required to sustain high growth rates, including nutrient transporters and metabolic proteins, which correlated with phenotypic differences between hiPSCs and hESCs. Thus, higher levels of glutamine transporters correlated with increased glutamine uptake, while higher levels of proteins involved in lipid synthesis correlated with increased lipid droplet formation. Some of the biggest metabolic changes were seen in proteins involved in mitochondrial metabolism, with corresponding enhanced mitochondrial potential, shown experimentally using high-resolution respirometry. hiPSCs also produced higher levels of secreted proteins, including ECM components and growth factors, some with known tumorigenic properties, as well as proteins involved in the inhibition of the immune system. Our data indicate that reprogramming of human fibroblasts to iPSCs effectively restores protein expression in cell nuclei to a state comparable to hESCs, but does not similarly restore the profile of cytoplasmic and mitochondrial proteins, with consequences for cell phenotypes affecting growth and metabolism. The data improve understanding of the molecular differences between induced and embryonic stem cells, with implications for potential risks and benefits for their use in future disease modelling and therapeutic applications.

eLife Assessment

This study reports differences in proteomic profiles of embryonic versus induced pluripotent stem cells. This **important** finding cautions against the interchangeable use of both types of cells in biomedical research, although the mechanisms responsible for these differences remains unknown. The proteomic evidence is **convincing**, even though there is limited validation with other methods.

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Introduction

Human embryonic stem cells (hESC) are derived from the inner cell mass of a pre-implantation embryo¹. They show prolonged undifferentiated potential, as well as the ability to differentiate into the three main embryonic germ layers², making them excellent models for studying disease mechanisms, development and differentiation. However, their use remains restricted by regulations, based in part upon ethical considerations³.

Over a decade ago, methods allowing the induction of pluripotent stem cells from fibroblast cultures, in both human and mice, were developed^{4,5}. These reports showed that by exogenously expressing a small set of key transcription factors (Oct4, Sox2, c-Myc and Klf4), a somatic cell could be reprogrammed back into a pluripotent state, characterised by their capacity for self-renewal and ability to differentiate into the three main germ layers. These human induced pluripotent stem cells (hiPSCs) show many key features of their physiological embryonic stem cell (hESC) counterparts, while avoiding many of the ethical issues regarding the use of stem cells derived from embryos.

Since the discovery of reprogramming methods, hiPSC lines have attracted great interest, particularly for their potential use as alternatives to hESCs in regenerative medicine⁶ and disease modelling, including studies on monogenic disorders^{7,8} and some late onset diseases⁹. However, to understand the value of using hiPSCs in regenerative therapy, drug development and/or studies of disease mechanisms, it is important to establish how similar hiPSCs are to hESCs at the molecular and functional levels. To address this, multiple studies have compared hiPSCs and hESCs, using a variety of assays, including methylation analysis¹⁰, transcriptomics^{11,12} and even quantitative proteomics^{13,14}. It should be noted, however, that many of these earlier studies were performed at a time when reprogramming protocols were less robust¹⁵ and when the depth of proteome coverage and quantitative information that could be obtained was lower than today.

In this study, we have addressed the similarity of hiPSCs to hESCs by performing a detailed proteomic analysis, comparing a set of four hiPSC lines derived from human primary skin fibroblasts¹⁶ of independent, healthy donors, with four independent hESC lines. The data highlight that while both types of stem cell lines have very similar global protein abundance profiles, they also show some specific and significant quantitative differences in protein expression. In particular, the reprogramed iPSC lines consistently display higher total protein levels, predominantly affecting cytoplasmic proteins required to sustain higher growth, along with mitochondrial changes, and an excess of secreted proteins, with impact upon cell phenotypes.

Results

hESCs and hiPSCs display quantitative differences in protein abundances

For this study, we compared multiple hESC and hiPSC lines, all derived from different donors and cultured using identical growth conditions. First, the expression levels of the main pluripotency markers were verified in each of the lines, with no differences detected between the respective hESC and hiPSC cell types (**Fig. 1a**). From these data, representative sets of four hiPSCs and four hESCs lines were selected for detailed proteomic analysis using mass spectrometry. The proteomes were characterised using tandem mass tags (TMT), within a single 10-plex (Table S1) and using MS3-based synchronous precursor selection (SPS). To further optimise quantification accuracy, each sample was allocated to a specific isobaric tag to minimise cross-population reporter ion interference (**Fig. 1b**), as previously described. In total 8,491 protein groups (henceforth referred to as ‘proteins’), were detected at 1% FDR, with >99% overlap between the proteins detected from both the hESC and hiPSC lines (**Fig. 1c**). However, it is important to highlight that TMT is not the best method to use when looking for proteins that are specific to one condition or population, as protein detected in one channel are frequently seen across all other channels as well.

To provide a quantitative comparison of the respective proteomes, we focussed on analysing the 7,878 proteins that were detected with at least 2 unique and razor peptides (see methods). After confirming that there were no differences in the abundance levels of histones between the two cell types (**Fig. S1**), protein copy numbers were estimated via the “proteomic ruler” (see methods). The copy number data (Supplemental Table 1) highlighted that both the hESC (**Fig. 1d**) and hiPSC (**Fig. 1e**), proteomes display a similar dynamic range, with estimated protein copy numbers extending from a median of less than 100 copies, to over 100 million copies per cell. Furthermore, the composition of the respective proteomes is highly similar. Both cell types display high expression levels of ribosomal proteins, protein chaperones and glycolytic enzymes (**Fig. 1f&g**), consistent with rapid proliferation and dependence on glycolysis for energy generation. It is only when the quantitative data are examined in more detail that differences between the cell types become apparent (**Fig. 1h**). A principal component analysis (PCA), based on the protein copy numbers, revealed a clear separation between the two stem cell populations within the main component of variation, which accounted for 69% of variance. The PCA suggested that the independent hiPSC lines were clearly different to the hESC lines, and vice versa.

Standard normalisation methods mask changes in total protein content in hiPSCs compared to hESCs

A previous proteomic study reported that there were virtually no protein level differences between hESCs and hiPSCs. However, in that study the intensity data were median normalised. We therefore decided to compare two different normalisation methods: i.e., the previously used median normalisation method and the “proteomic ruler”. The median normalisation produces concentration-like results and is frequently used to normalise proteomic data. With this approach, our data also show no major differences in protein abundances between the hESC and hiPSC lines (**Fig. 2a**; Supplemental Table 2), i.e. ~94% of all proteins displayed no significant changes in abundance ($FC > 1.5$ -fold; q -value < 0.001), similar to the previously reported conclusion. However, median (or total intensity) normalisation methods lack the capacity to detect changes in absolute abundance, cell size or protein content. By artificially forcing all medians to be almost identical, such changes are invisible.

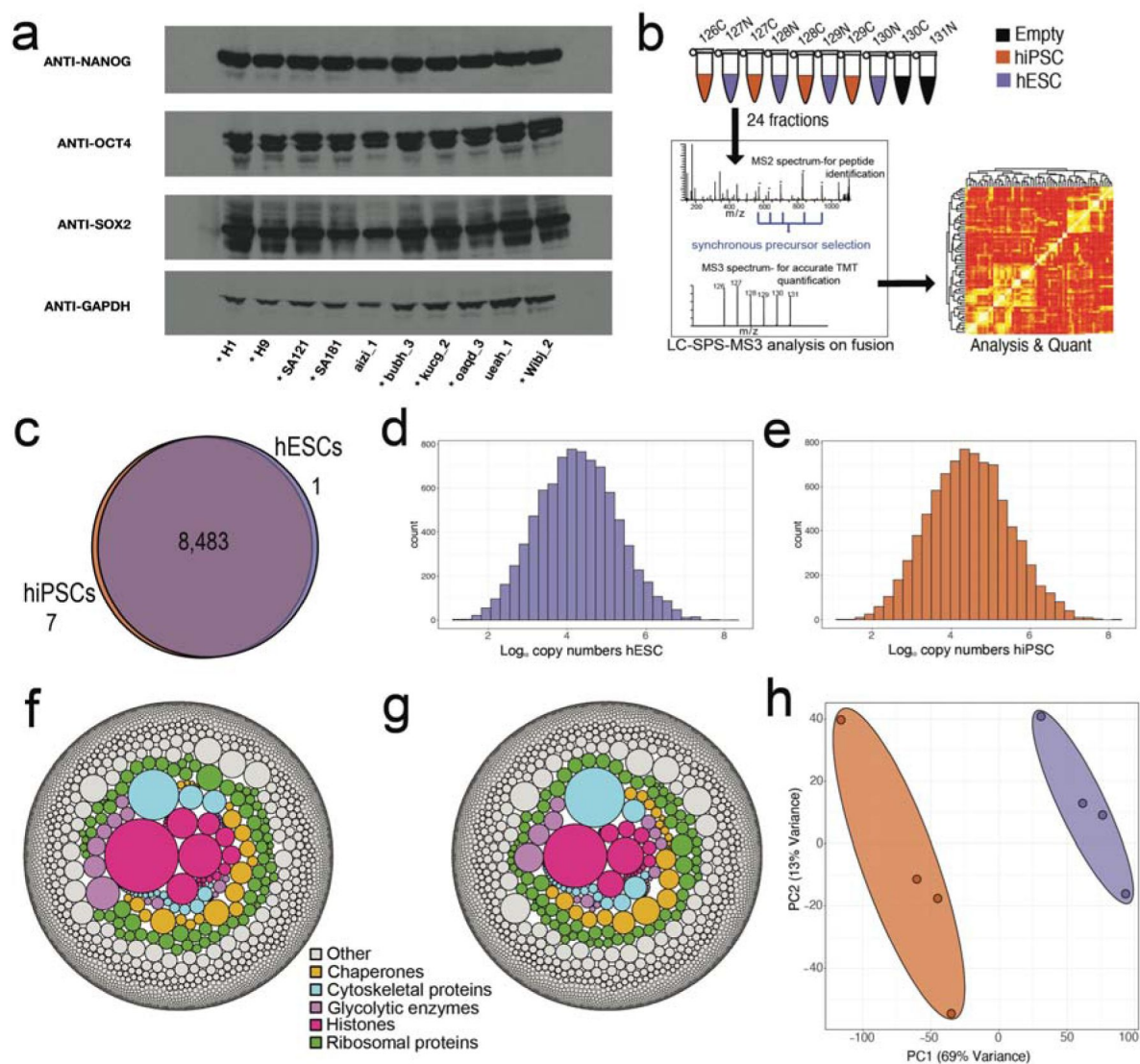


Figure 1-

Proteomic overview:

(a) Western blots showing the expression of the pluripotency factors NANOG, OCT4 and SOX2 across all hESC and hiPSC lines. The 8 lines showed with * were used within the proteomic analysis (b) Diagram showing the SPS-MS3 TMT proteomic workflow used for the experiment. (c) Venn diagram showing the overlap of proteins identified within the hiPSC and hESC populations. (d) Average copy number histogram for the hESCs. (N=4) (e) Average copy number histogram for the hiPSCs (N=4). (f) Bubble plot showing proteins coloured by specific categories where the size is represented by the average hESC estimated protein copy numbers. (g) Bubble plot showing proteins coloured by specific categories where the size is represented by the average hiPSC estimated protein copy numbers. (h) PCA plot based on the log₁₀ copy numbers for all 8 stem cell lines. hESCs are shown in purple and hiPSCs in orange.

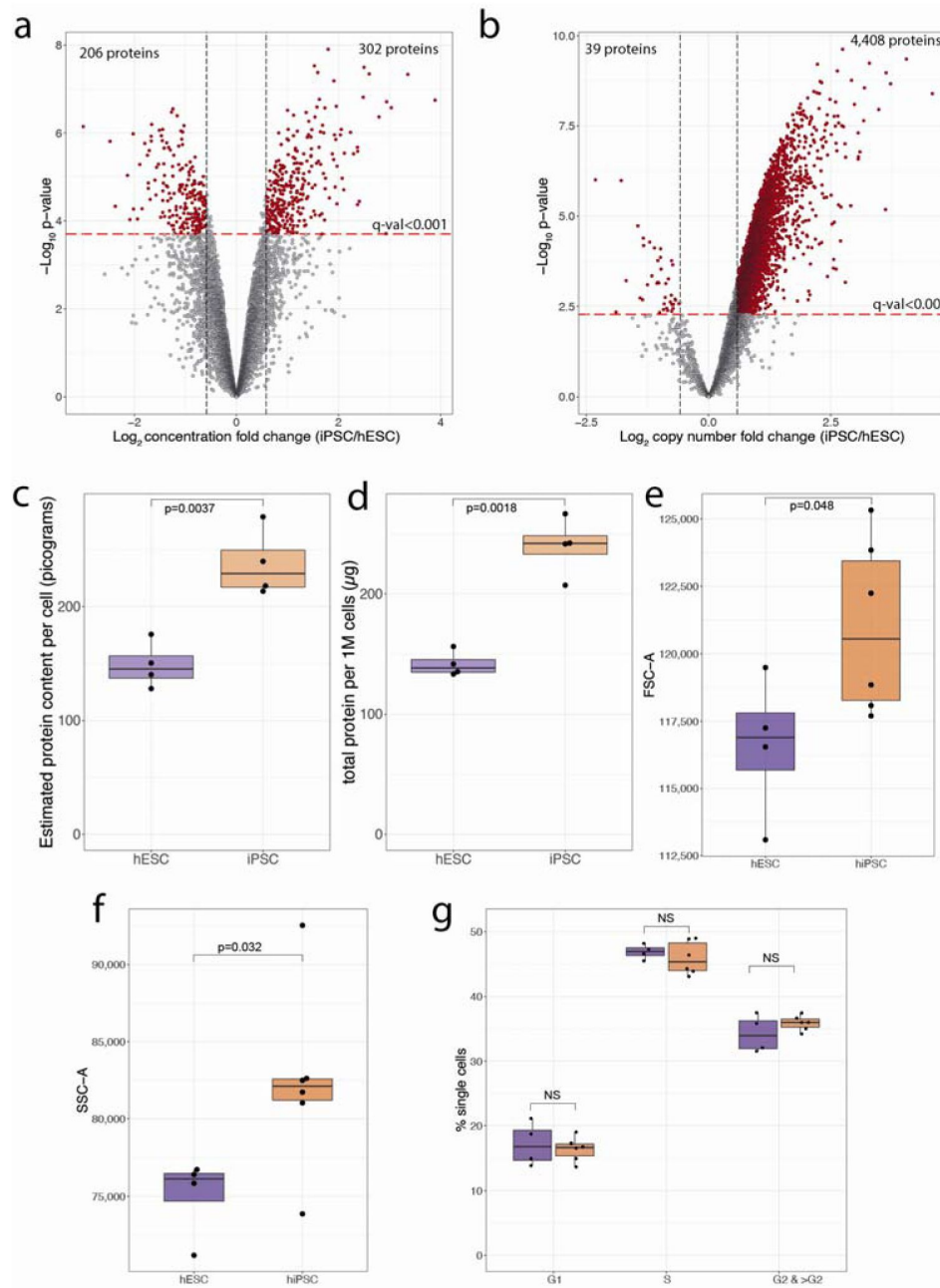


Figure 2-

Normalisation and protein content:

(a) Concentration based volcano plot showing the $-\log_{10} p\text{-value}$ and the $\log_2$ fold change comparing hiPSCs (N=4) to hESCs (N=4). Elements shaded in red are considered significantly changed. All dots above the red line have a q-value lower than 0.001 **(b)** Copy number-based volcano plot showing the $-\log_{10} p\text{-value}$ and the $\log_2$ fold change comparing hESCs (N=4) to hiPSC (N=4). Elements shaded in red are considered significantly changed. All dots above the red line have a q-value lower than 0.001 **(c)** Box plot showing the MS based estimated protein content for hESCs (N=4) and hiPSC (N=4). **(d)** Box plot showing the protein amount per million cells derived from the EZQ Protein Quantification Kit for all hESCs (N=4) and hiPSC (N=4) **(e)** Boxplot showing the median forward scatter of hESCs (N=4) and hiPSCs (N=6). **(f)** Boxplot showing the median side scatter of hESCs (N=4) and hiPSCs (N=6). **(g)** Boxplot showing the median percentage of cells across cell cycle stages for hESCs (N=4) and hiPSCs (N=6). For all boxplots, the bottom and top hinges represent the 1st and 3rd quartiles. The top whisker extends from the hinge to the largest value no further than 1.5 3 IQR from the hinge; the bottom whisker extends from the hinge to the smallest value at most 1.5 3 IQR of the hinge.

This is not the case for the results produced with the “proteomic ruler”²⁰. The copy number-based analysis enables an approximation to absolute protein abundance and can reveal changes in cell mass, as we previously reported^{22,23}. Using the proteomic ruler method highlighted systematic differences between hESCs and hiPSCs (**Fig. 2b**; Supplemental Table 3), with 56% (4,426/7,878) of all proteins detected significantly increased in hiPSCs (FC>1.5-fold; q-value < 0.001) and with particular enrichment in translation-related processes (Fig. S2). In contrast, only 40 proteins (0.5%) showed significantly lower expression levels in hiPSCs. With thousands of proteins displaying higher abundance, we hypothesised that hiPSCs have higher total protein content, compared to hESCs. Using the protein copy numbers to estimate the total protein content showed that hiPSCs had >50% higher protein content compared to hESCs (**Fig. 2c**). To validate this observation, an independent assay (EZQTM assay; see methods), was used to measure the total protein yield from similar numbers of freshly grown hiPSCs and hESCs. From these experiments, the calculated protein amount per million cells was >70% higher (**Fig. 2d**; p-value=0.0018) in hiPSCs, relative to hESCs. We conclude that hiPSCs have a higher total protein content. Changes in protein content could potentially be linked to differences in the cell cycle profile. Hence, to test this, we used fluorescence-activated cell sorting (FACS) to study the cell cycle distribution of hESCs and hiPSCs. The FACS data showed that hiPSCs have significantly higher forward scatter (**Fig. 2e**), correlated to increased cell size, as well as significantly higher side scatter (**Fig. 2f**), correlating to increased cell granularity. However, the FACS analysis revealed no significant differences between hiPSCs and hESCs in the percentage of cells at each of the cell cycle stages (**Fig. 2e**). We conclude that hiPSCs have significantly higher total protein content, with increased size and granularity, but that these differences with hESCs are independent of changes in cell cycle distribution.

hiPSCs have elevated nutrient transporters, metabolic proteins, and protein and lipid synthesis machinery

To maintain a higher protein content than hESCs with a comparable cell cycle profile, hiPSCs would require higher protein synthesis capacity, which in turn requires increased access to nutrients and energy. Energy metabolism in primed pluripotent stem cells is largely dependent on glycolysis²⁴, which is sensitive to glucose uptake and lactate shuttling. Therefore, we compared the expression of the respective glucose and lactate transporters between hiPSCs and hESCs. The data showed both main glucose transporters, GLUT1 (SLC2A1) and GLUT3 (SLC2A3), had higher abundance in hiPSCs, as did the lactate transporters SLC16A1 and SLC16A3 (**Fig. 3a**). Other rate limiting enzymes, including Hexokinase 1 (HK1) and 2 (HK2) were also significantly increased within hiPSCs (**Fig. 3b**), suggesting increased glycolytic potential.

Nutrient uptake is mostly handled by the SLC (solute carrier), group of membrane transporters. Analysis of the 15 SLC transporters that are most upregulated in hiPSCs, compared to hESCs, showed that they mostly belonged to two categories, i.e., amino acid and mitochondrial transporters (**Fig. 3c**). Amino acids are vital to sustain high rates of protein synthesis²⁵ and the data showed that 11/12 amino acid transporters were significantly increased in hiPSCs, compared to hESCs, including the hyper abundant protein SLC3A2, which is present at >4 million copies per cell (**Fig. 3d**). The highest fold increases, (>4-fold), were seen for SLC38A1 and SLC38A2, both of which are major glutamine transporters^{26,27}. We next examined whether the increased abundance of the glutamine transporters had phenotypic impact, i.e., whether it correlated with increased glutamine uptake within hiPSCs. To test this hypothesis, we measured the uptake of radio-labelled glutamine in both hiPSCs and hESCs (see methods). The data showed that hiPSCs had a median of >90% higher uptake of glutamine, compared to hESCs (**Fig. 3e**). Glutamine has been reported to be the most consumed amino acid in hESCs²⁸ and its catabolism to be one of the vital metabolic pathways that can provide ATP and more importantly biosynthetic precursors, required to sustain growth²⁹. Hence, we also explored the abundance of enzymes involved in glutaminolysis and found that vital proteins, including GLS, GLUD1, GPT2 and GOT2, were also significantly higher in hiPSCs (**Fig. 3f**).

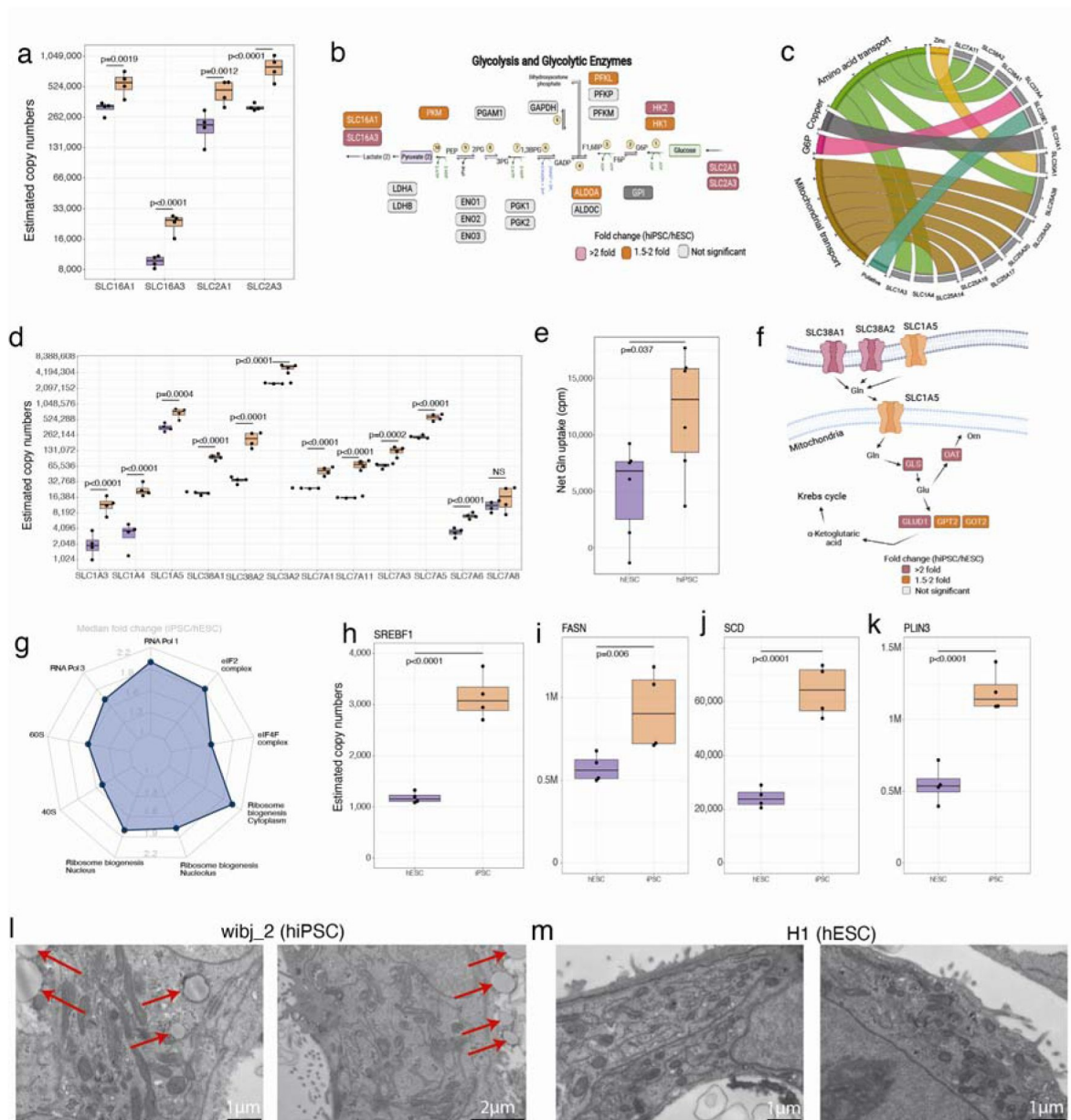


Figure 3-

Fuelling growth:

(a) Boxplots showing the estimated copy numbers for the lactate (SLC16A1 and SLC16A3) and glucose transporters (SLC2A1 and SLC2A3) across hESCs (N=4) and hiPSC (N=4). **(b)** Schematic showing the glycolytic proteins and their fold change in hESC vs hiPSCs. **(c)** Chord diagram showing the 15 most upregulated solute carrier proteins along with their classification based on transport activities/localisation. **(d)** Boxplots showing the estimated copy numbers of the main amino acid transporters in hESCs (N=4) and hiPSC (N=4). **(e)** Boxplot showing the net glutamine uptake (see methods) in hESCs (N=4) and hiPSC (N=4). **(f)** Schematic showing the glutaminolysis proteins and their fold change in hESCs (N=4) and hiPSC (N=4). **(g)** Radar plot showing the median fold change (hiPSC/ESC) for protein categories which are related to the pre-ribosomes. Boxplots showing the estimated copy numbers for **(h) SREBF1**, **(i) FASN**, **(j) SCD**, **(k) PLIN3**, in hESCs (N=4) and hiPSC (N=4). **(l)** Transmission electron microscopy images for wibj_2 (hiPSC). Lipid droplets are marked with red arrows. **(m)** Transmission electron microscopy images for H1 (hESC). For all boxplots, the bottom and top hinges represent the 1st and 3rd quartiles. The top whisker extends from the hinge to the largest value no further than 1.5 3 IQR from the hinge; the bottom whisker extends from the hinge to the smallest value at most 1.5 3 IQR of the hinge.

Having established that hiPSCs have increased expression of nutrient transporters and higher expression of enzymes in key metabolic pathways, compared with hESCs, we next looked at the machinery required for protein synthesis. The levels of many of the proteins involved in ribosome subunit biogenesis, including ribosomal proteins, were higher in hiPSCs (**Fig. 3g**). The increased expression of translation machinery components, nutrient transporters and many metabolic enzymes, is consistent with the increased total protein content seen within hiPSCs.

The data also highlighted increased fatty acid (FA) and lipid droplet (LD) synthesis potential in hiPSCs, with increased abundance of the protein SREBP1 (SREBF1; **Fig. 3h**), a master regulator of lipid synthesis³⁰, as well as FASN (**Fig. 3i**) and SCD (**Fig. 3j**). Similarly, a crucial regulator for LD assembly, PLIN3³¹, (**Fig. 3k**), displayed >2-fold increased abundance in hiPSCs. To examine the potential phenotypic impact of this increased abundance of proteins involved in LD synthesis, we performed transmission electron microscopy (TEM) analyses to compare hiPSC and hESC cells. This showed that LDs were clearly visible in hiPSCs (**Fig. 3l**), but not visible in hESCs (**Fig. 3m**). We conclude that the hiPSCs have elevated levels of LDs, resulting from the increased expression of proteins involved in lipid synthesis and LD assembly.

hiPSCs show altered mitochondrial metabolism compared to hESCs

Our data also highlighted important changes in mitochondrial proteins, including increases in the levels of metabolic proteins that are encoded within the mitochondrial genome³², (**Fig. 4a**). The latter proteins, which are translated by specialised mitochondrial ribosomes (mitoribosomes), are embedded in the mitochondrial membrane. The protein components of mitoribosomes also showed increased expression in hiPSCs (**Fig. 4b**), along with virtually all proteins involved in the translation initiation, elongation and termination of mitochondrial genome-encoded proteins (**Fig. 4c**).

The analysis of transporter proteins revealed a cluster of 22/27 mitochondrial transporters that were significantly increased in hiPSCs, including the hyper abundant, (>10 million copies per cell), ATP/ADP transporter (**Fig. 4d**). A subset of 14 transporters displayed >2-fold increased abundance, including the acylcarnitine transporter SLC25A20 (**Fig. 4e**), which is part of the carnitine shuttle in the beta oxidation pathway. Another component of the carnitine shuttle, CPT1A, displayed over 4-fold higher abundance in hiPSCs (**Fig. 4f**), suggesting an important role. The data showed it was not just fatty acid oxidation, but also synthesis, that was affected, with proteins acting in the mitochondrial fatty acid synthesis (mFAS) pathway also increased in abundance. MCAT (**Fig. 4g**), MECR (**Fig. 4h**) and OXSM (**Fig. 4i**), all displayed ~2-fold higher abundance in hiPSCs compared to hESCs. These results have a metabolic relevance as mFAS has been reported to control the activity of the electron transport chain (ETC)³³, which was also increased in hiPSCs, with subunits of all 5 ETC complexes increased in abundance in hiPSCs and with complex II and complex III showing the most prominent effects (**Fig. 4j**). Complex II is also part of the tricarboxylic acid (TCA) cycle, which displayed increased abundance of the majority of proteins involved in the pathway (**Fig. 4k**).

As the proteomic data showed clear differences between hiPSCs and hESCs in the levels of mitometabolism proteins, we performed experiments to explore whether this was reflected in phenotypic differences between hiPSCs and hESCs. This was tested using high-resolution respirometry (see Methods). The data showed that hiPSCs had a higher P/E control ratio to hESCs, which denotes an increased capacity of the phosphorylation system to produce ATP (**Fig. 4l**). We conclude that hiPSCs have elevated levels of mitometabolism proteins relative to hESCs, resulting in higher respiratory activity.

hiPSCs upregulate secreted proteins affecting their microenvironment

Among the most upregulated proteins in hiPSCs were a subset of secreted proteins. Secreted proteins are of great importance because changes in their absolute abundance can affect the extracellular environment. These secreted proteins mostly represented 4 categories: structural extracellular matrix (ECM) proteins, growth factors, protease inhibitors and proteases (**Fig. 5a**). The ECM both provides physical support for cells, as well as actively participating in cell signalling by providing domains for growth factors³⁴. The ECM can also be reshaped in tumours, thereby promoting cancer cell growth and migration³⁵. The current data show that both laminins and collagens were all increased in abundance in hiPSCs (**Fig. 5b**). Collagens are reported to alter the stiffness of the ECM and their synthesis is iron intensive. Interestingly, the data also show that proteins involved in importing and storing iron were increased in abundance in hiPSCs (**Fig. 5c**).

The data also showed that 13 growth factors were increased in abundance in hiPSCs, compared to hESCs. A subset of these, i.e., FGF1, FGF2 and NODAL, are reported to have direct relevance to the maintenance of pluripotency and can modulate important processes in PSCs^{36–38} (**Fig. 5g**). Other growth factors that are upregulated in hPSCs are linked to disease mechanisms and cancer. This includes VGF (**Fig. 5h**), which is linked to promoting growth and survival in glioblastoma³⁹ and MDK (**Fig. 5i**), which is highly expressed in malignant tumors⁴⁰ and has been shown to play a role in chemoresistance⁴¹.

hiPSCs display increased abundance of immunosuppressive proteins

NODAL wasn't the only growth factor in the TGFB family that was increased in hiPSCs, with TGFB1 displaying a ~5-fold increase in abundance in hiPSCs compared to hESCs (**Fig. 5j**). Besides its role as a growth factor, TGFB1 has been shown to have important roles in the regulation of the immune response, promoting the generation of regulatory T cells, while inhibiting the generation and function of effector T cells. As immunogenicity of PSCs is a topic of relevance to clinical adaptations, we looked for differences in modulators of the immune response.

Arginine availability is vital to effector T cells and other leukocytes, where depletion mediated by Arginase has been shown to be linked to T cell inhibition⁴². Our data show that hiPSCs have ~2.5-fold higher abundance of ARG1 (**Fig. 5k**). Furthermore, hiPSCs also display increased expression of the immune checkpoint protein, CD276 (**Fig. 5l**), which has been reported to be a potent inhibitor of survival and function of T cells^{43,44}.

hiPSCs also displayed increased abundance of inhibitory ligands that suppress the immune function of other leukocytes. The data show hiPSCs have increased abundance of the non-classical HLA-E (**Fig. 5m**), which has been shown to interact with the NK cell receptor, NKG2A, to mediate immune evasion in ageing cells⁴⁵. They also displayed increased abundance of CD200 (**Fig. 5n**), a ligand for CD200R, which can inhibit the immune response from macrophages, basophils, NK cells and T cells, as well as CD47 (**Fig. 5o**), a ligand of SIRPA that helps cells to escape macrophage phagocytosis. These data indicate that hiPSCs have increased abundance of known immunosuppressive proteins, compared to hESCs.

hiPSCs display reduced abundance of H1 histones

A striking feature of this proteomic study is how few proteins (<1%; 40/7,878), showed significantly decreased abundance in hiPSCs, compared to hESCs. A high proportion of these decreased abundance proteins affect nuclear processes. Thus, an overrepresentation analysis showed that proteins whose abundance was decreased in hiPSCs were enriched in GO terms related to DNA

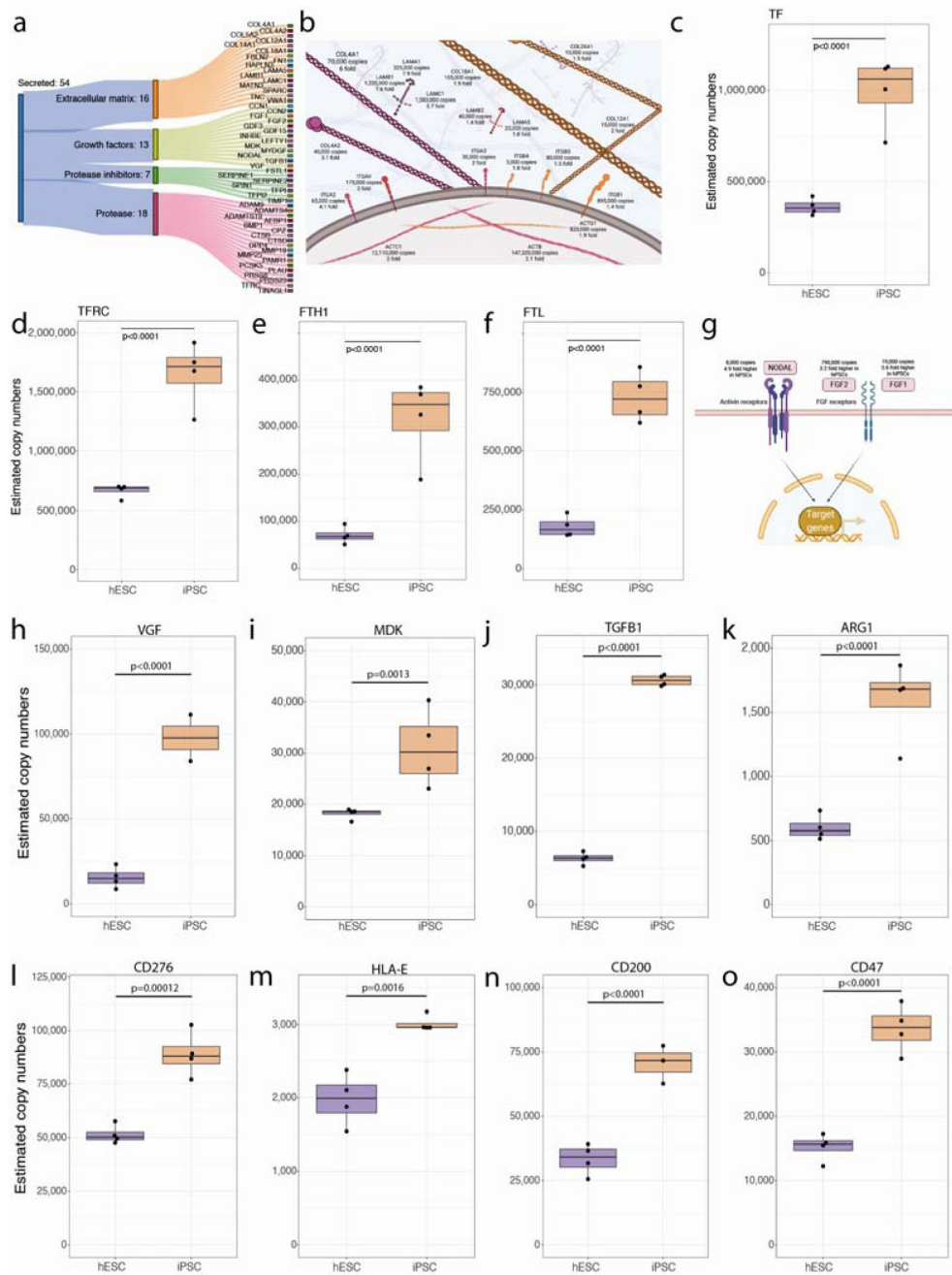


Figure 5-

Secreted proteins:

(a) Sankey diagram showing the secreted proteins that belong to the extracellular matrix (ECM), Growth factor, Protease Inhibitor or Protease categories and are significantly increased in abundance in hiPSCs (b) Schematic showing ECM proteins that are significantly increased in abundance in hiPSCs. Boxplot showing the estimated copy numbers for (c) TF, (d) TFRC, (e) FTH1, and (f) FTL in hESCs (N=4) and iPSC (N=4). (g) Schematic showing the changes in abundance in primed pluripotency growth factors. (h) Boxplot showing the estimated protein copy numbers for VGF. (i) Boxplot showing the estimated protein copy numbers for MDK. Boxplot showing the estimated protein copy numbers for (j) TGFB1, (k) ARG1, (l) CD276, (m) HLA-E, (n) CD200 and (o) CD47. All boxplots show the data for hESCs and iPSCs. For all boxplots, the bottom and top hinges represent the 1st and 3rd quartiles. The top whisker extends from the hinge to the largest value no further than 1.5 IQR from the hinge; the bottom whisker extends from the hinge to the smallest value at most 1.5 IQR of the hinge.

recombination, nucleosome positioning and chromatin silencing (**Fig. 6a**). Notably, this included four H1 histone variants, which are reported to influence nucleosomal repeat length⁴⁶ and stabilise chromatin structures⁴⁷. Our data show that the most abundant H1 variant in hESCs, HIST1H1E, is decreased in abundance in hiPSCs by ~3.5-fold (**Fig. 6b**), while HIST1H1C (**Fig. 6c**), HIST1H1D (**Fig. 6d**) and H1FX (**Fig. 6e**) are all decreased by >1.7-fold.

As histone variants have very similar protein sequences, where peptides detected by MS can potentially match to multiple H1 histone variants, a peptide level analysis was necessary to deconvolute the signal (**Fig. S3**). The Andromeda search engine⁴⁸ assigns peptide intensities to a protein following a razor peptide approach, where the intensity of a peptide is assigned to only one protein, regardless of whether it is unique, or shared by two or more proteins. This makes the analysis of specific variants challenging at the protein level. Hence, we focussed on a peptide-specific approach with the MS data and found that the intensity of the peptides that were shared between these 3 H1 histone variants displayed a consistent reduction in abundance in hiPSCs (**Fig. S3**).

The systematic reduction in abundance in hiPSCs seen with H1 histone variants was not seen for members of the other histone families. Evaluating either the concentration (**Fig. S1**), or copy numbers (**Fig. 6f**), across all histones, showed no significant differences in expression between hiPSCs and hESCs. Furthermore, for core histones, including H3 and H4, there were no significant abundance differences seen within either the proteomics data (**Fig. 6g&h**), or in additional western blot analyses that were performed to validate these conclusions (**Fig. 6i&j**). However, we did detect differences between hiPSCs and hESCs in the expression of histone H2 variants, with H2AFV (**Fig. 6k**), H2AFY (**Fig. 6l**) and H2AFY2 (**Fig. 6m**), all increased in abundance in hiPSCs. As histone H2 variants also have high sequence similarity and shared peptides, we also performed a peptide level analysis, which validated that both shared and unique peptides displayed the same pattern, i.e., showing increased abundance in hiPSCs (**Fig. S4**). Thus, we conclude that there are opposing effects for histone H1 and histone H2 variants, with the former decreased and the latter increased in abundance in hiPSCs.

Discussion

Induced pluripotent stem cells can provide valuable models for clinical research and future therapies, which makes it vital to understand both their similarities and any specific differences, with embryo-derived human stem cells. This study provides a detailed comparison of the proteomes of multiple hiPSC and hESC lines derived from different donors. The major conclusion is that while hiPSC and hESC lines express a near identical set of proteins, with similar abundance ranks, they also display important quantitative differences. In particular, our data indicate that hiPSCs reprogrammed from skin fibroblasts display considerable differences in their cytoplasmic and mitochondrial proteomes, compared to hESCs, while the nuclear proteome was very similar between the two cell types. Furthermore, additional microscopy analyses and functional assays showed that the systematic differences in the proteomes of the respective hiPSCs and hESCs had a measurable impact on cell phenotypes, most notably affecting mitochondria, metabolic activity, and nutrient transport. It would be of interest in future to analyse whether hiPSCs reprogrammed from different cell types, such as peripheral leukocytes, also share these differences in protein expression with hESCs, or if these specific changes are characteristic of fibroblast-derived iPS cells.

Using estimated protein copy numbers, our data show that only <1% of proteins were significantly decreased in abundance in hiPSCs, compared to hESCs, including multiple H1 histones. In contrast, ~56% of all proteins quantified were significantly increased in hiPSCs (fold change>1.5 and q-value <0.001), with most of these increases affecting cytoplasmic and mitochondrial proteins and activities. The MS data show that total protein levels are higher overall in hiPSCs, as compared with hESCs, a result that was independently validated using an EZQ protein assay. This difference

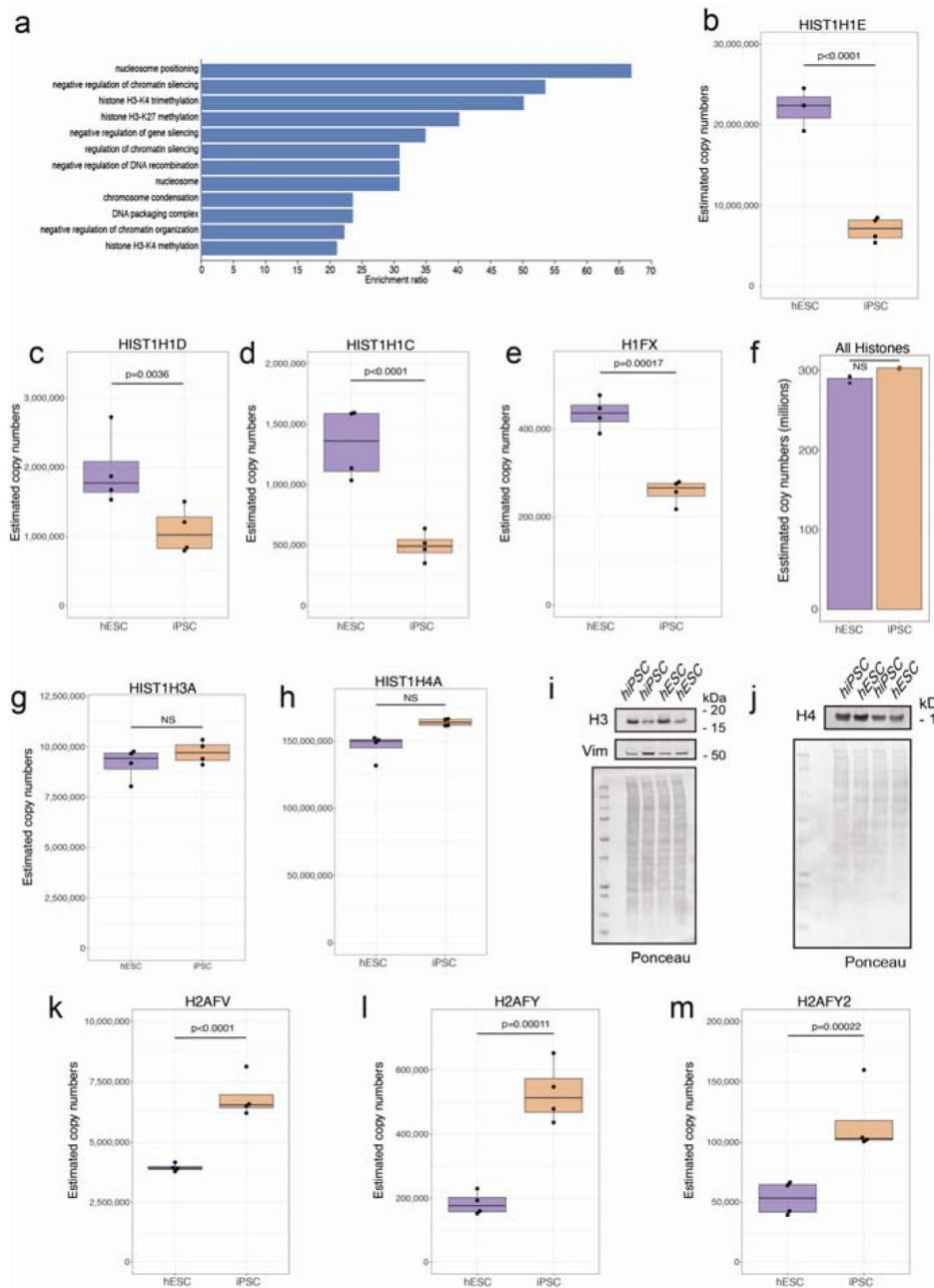


Figure 6 –

Changes within histones:

(a) Barplot showing the GO term enrichment results for proteins significantly decreased in abundance (see methods) in hiPSCs. Boxplots showing estimated copy numbers for (b) HIST1H1E, (c) HIST1H1D, (d) HIST1H1C and (e) H1FX in hESCs (N=4) and hiPSC (N=4). (f) Barplot showing the median estimated copy numbers for all histones in hESCs and hiPSCs. Boxplots showing estimated copy numbers for (g) HIST1H3A in and (h) HIST1H4A in hESCs and hiPSCs. Western blot showing the abundance of (i) H3 and (j) H4 histones in hESCs (N=4) and hiPSC (N=4). Boxplots showing estimated copy numbers for (k) H2AFV, (l) H2AFY and (m) H2AFY2 in hESCs (N=4) and hiPSC (N=4). For all boxplots, the bottom and top hinges represent the 1st and 3rd quartiles. The top whisker extends from the hinge to the largest value no further than 1.5 3 IQR from the hinge; the bottom whisker extends from the hinge to the smallest value at most 1.5 3 IQR of the hinge.

in total protein content was shown by FACS analysis not to result from systematic differences between hiPSCs and hESCs in cell cycle progression. Instead, the increased protein levels in hiPSCs correlated with increased levels of the protein translational machinery, along with increased metabolic and mitochondrial activity and higher levels of nutrient transport.

These results highlight an important technical point relating to data normalisation and its effect on the interpretation of such data. By using a standard median normalisation (concentration-based approach), instead of the proteomic ruler²⁰, the difference in total protein content between the cell types, involving the increased abundance of thousands of proteins, is not apparent. Hence, two cell types with, for example, a 4-fold higher total protein content, could nonetheless appear to show virtually no significant differences, as long as the protein ranks and relative concentrations remained similar. This would result in an erroneous conclusion that there is little to no change in protein expression, while the orthogonal data suggest otherwise.

Having established that hiPSCs displayed higher total protein content than corresponding hESC lines, we sought to understand how this could be maintained. To maximise protein synthesis, nutrient availability and energy production are key factors²⁵. The proteomic data show that vital nutrient transporters, known to be important for growth and protein production⁴⁹, were significantly increased in hiPSCs, compared to hESCs. In particular, the 3 glutamine transporters, SLC1A5, SLC38A1 and SLC38A2, were all significantly increased in abundance, with additional functional assays showing that this correlated with higher levels of glutamine uptake measured in hiPSCs. Glutamine has been previously shown to fuel growth and proliferation in rapidly dividing cells, including cancer cells²⁶ and could be sustaining higher rates in hiPSCs.

Nutrients provide the fuel, but it is the metabolic proteins that are the engines that convert them to energy. Here, our data showed that proteins involved in both glycolysis and glutaminolysis were significantly increased in abundance in hiPSCs. When cells preferentially use the glycolytic pathway, e.g., stem cells and cancer cells, there is increased demand for biosynthetic precursors and NADPH⁵⁰. These precursors can be supplied via glutaminolysis^{51–53} linked to the TCA, both important mitochondrial processes and both with significantly increased protein levels in hiPSCs, along with other proteins involved in the electron transport chain. One caveat is that an uneven distribution of biological sex between donors of the respective hESC and hiPSCs lines, could mean there is a potential for sex-specific differences to explain at least some of these effects, however in previous a large-scale proteomic study of male and female hiPSCs no changes in proteins involved in oxidative phosphorylation were detected⁵⁴. Differences in the mitochondria between hiPSCs and hESCs have been previously reported, but whether they originate from the reprogramming process, or are induced by the increased nutrient uptake, remains unknown and is a point of interest for future studies.

Secreted proteins, such as growth factors and ECM proteins, are a category of great interest, because their absolute abundance can affect the surrounding cellular microenvironment. hiPSCs were found here to show increased expression levels of growth factors that are linked to cancer and immunosuppression. For example FGF2, an important growth factor for primed pluripotent stem cells, has been shown that to promote ERK activation⁵⁵, stimulating protein synthesis^{56–58}. Thus, the increased abundance of FGF2 could generate a feedforward loop, further driving/sustaining growth in hiPSCs, however that growth potential is also linked to breast⁵⁹ and gastric cancers⁶⁰, as well as gliomas⁶¹. Another important growth factor that is increased in abundance in hiPSCs is TGFB1, a known potent inhibitor of T cell responses^{62,63}. We note that the immunogenicity of pluripotent stem cells has important consequences for cell therapy applications. Our data suggest that hiPSCs might have a higher immune evasion potential, via multiple mechanisms. They display increased abundance of secreted T cell inhibitors, including TGFB1 and ARG1⁶⁴, along with inhibitory ligands, such as CD276, CD200 and CD47. An increased inhibitory capacity, combined with the tumorigenic potential of hiPSCs⁶⁵, raises some concerns about the suitability of using reprogrammed hiPSCs for certain types of therapeutic applications.

Based upon our current data, we recommend that potential phenotypic consequences resulting from the observed differences in iPSC and ESC proteomes should be studied further to determine whether the immunosuppressive properties of these cells vary significantly.

In summary, our data show that hiPSCs and hESCs, despite their clear similarities, are not identical at both the protein and phenotypic levels. We show that hiPSCs reprogrammed from skin fibroblasts differ from hESCs, predominantly in their cytoplasmic and mitochondrial proteome, leading to measurable functional differences affecting their metabolic activity and growth potential. These data can help to inform future strategies to mitigate for these differences as hiPSCs continue to be used in important clinical applications and as disease models.

Materials and methods

HipSci hiPSC line generation

As part of the HipSci project¹⁶, hiPSC lines were generated in the sanger centre. hiPSCs were generated from fibroblasts obtained by skin punch biopsies. The fibroblasts were reprogrammed as described previously¹⁶, in brief Sendai vectors expressing hOCT3/4, hSOX2 and hc-MYC were used.

HipSci hiPSC line quality control

The hiPSC lines were passaged a mean of 16 times before being subjected to the first tier of molecular data for quality control, which included genotyping ('gtarray'), gene expression data ('gexarray'), and an assessment of the pluripotency and differentiation potential of each line ('Cellomics'). Pluripotency of the lines was additionally verified *in silico*, using the PluriTest assay⁶⁶. Subsequently one or two lines per donor were subjected to a set of molecular data QC assays. The criteria for line selection were: (i) level of pluripotency, as determined by the PluriTest assay (ii) number of copy number abnormalities and (iii) ability to differentiate into each of the three germ layers. These included proteomics, DNA methylation ('mtarray'), RNA-sequencing and high-content cellular imaging.

hiPSC and hESC Cell Culture

Human iPSC cells generated by the HipSci consortium¹⁶ (aizi_1, bubh_3, kucg_2, oaqd_3, ueah_1 and wibj_2) and human hESCs (SA121 and SA181, H1, H9) were both grown in identical conditions, maintained in TERS medium⁶⁷ supplemented with FGF2 (Peprotech, 30 ng/ml) and noggin (Peprotech, 10 ng/ml) on growth factor reduced geltrex basement membrane extract (Life Technologies, 10 µg/cm²) coated dishes at 37°C in a humidified atmosphere of 5% CO₂ in air.

Cells were routinely passaged twice a week as single cells using TrypLE select (Life Technologies) and replated in TERS medium that was further supplemented with the Rho kinase inhibitor Y27632 (Tocris, 10 µM) to enhance single cell survival. Twenty-four hours after replating Y27632 was removed from the culture medium. For proteomic analyses cells were plated in 100 mm geltrex coated dishes at a density of 5×10⁴ cells cm⁻² and allowed to grow to for 3 days until confluent with daily medium changes.

Immunoblotting

Equal volumes of hiPSC or hESCs protein lysates were boiled in LDS/RA buffer for 5 mins at 95°C and loaded into 4-15% NuPAGE Bis-Tris SDS-PAGE gels in running buffer (50 mM MES, 50 mM Tris, 0.1 % SDS, 1 mM EDTA, pH7.3), transferred onto nitrocellulose membrane (Amersham #10600041) in transfer buffer (8 mM Tris, 30 mM Glycine, 20 % Methanol) and stained with Ponceau S (Sigma-Aldrich, #P7170). Membranes were blocked in TBS-T + 5% BSA for 1 hr at RT and incubated

overnight at 4°C in primary antibodies prepared in TBS-T + 5% BSA. Membranes were washed 3 x 15 mins in TBS-T, incubated with secondary antibody for 1 hr at RT, washed, and imaged using Odyssey CLx (LI-COR). Antibodies: Histone H3 (Abcam, ab1791, 1:1000); Histone H4 (Abcam, ab10158, 1:1000), Vimentin (Cell Signalling Technology (CST), #5741S, 1:1000), IRDye® 680RD Donkey anti-Rabbit IgG Secondary Antibody (LI-COR, 926-68073, 1:10,000), GAPDH (CST, #97166, 1:10000), OCT4A (CST, #2840, 1:10000), SOX2 (CST, #3579, 1:10000), NANOG (CST, #4903, 1:10000).

Flow cytometric analysis

Cells were seeded on geltrex coated dishes in TESR medium at a density of 3×10^3 cells/cm². After 24 hours the medium was replaced with fresh TESR medium and after a further 24 hours the cells were harvested using TrypLE select, resuspended in TESR medium and counted (cell density at harvest was approximately 1×10^5 cells/cm² for each cell line). 5×10^5 cells were then collected by centrifugation at 300 xg for 2 minutes then resuspended with 1ml of Dulbecco's PBS (without calcium or magnesium) containing 1% fetal bovine serum. The cells were then collected by centrifugation at 300 xg for 2 minutes and resuspended in 1ml of ice cold 90% methanol / 10% dH₂O while vortexing. Samples were then incubated for 30 minutes at room temperature before being stored at -20°C until they were analysed.

For cell cycle analysis the cells were collected by centrifugation at 300 xg for 2 minutes, washed with PBS, then resuspended in 300 micro l of staining buffer (Dulbecco's PBS + 1% FBS + 50 micro g/ml propidium iodide, 50 micro g/ml ribonuclease A) for 20 minutes at room temperature in the dark. Cellular DNA content was determined by analysis on a FACS Canto flow cytometer (BD Biosciences). PI fluorescence was detected using 488nm excitation and fluorescence emission collected at 585/42nm. Data was analysed using Flowjo software. Doublet discrimination was performed on the basis of PI-A v PI-W measurements and cell cycle distribution determined using the Watson Pragmatic model.

Cell line selection for mass spectrometry

Human iPS cells (bubh_3, kucg_2, oaqd_3 and wibj_2) and human hESCs (SA121 and SA181, H1 and H9) were analysed by mass spectrometry using TMT as described below.

Protein extraction

Cell pellets were resuspended in 300 µL extraction buffer (4% SDS in 100 mM triethylammonium bicarbonate (TEAB), phosphatase inhibitors (PhosSTOP™, Roche)). Samples were boiled (15 min, 95 °C, 350 rpm) and sonicated for 30 cycles in a bath sonicator (Bioruptor® Pico bath sonicator, Diagenode, Belgium; 30s on, 30s off) followed by probe sonication for 50 s (20s on, 5s off). 2 µL Benzonase® nuclease HC (250 U/µL, Merck Millipore) was added and incubated for 30 min (37 °C, 750 rpm). Reversibly oxidized cysteines were reduced with 10 mM TCEP (45 min, 22 °C, 1,000 rpm) followed by alkylation of free thiols with 20 mM iodoacetamide (45 min, 22 °C, 1,000 rpm, in the dark). Proteins were quantified using the fluorometric EZQ™ assay (Thermo Fisher Scientific).

Protein digestion using the SP3 method

Protein extracts were cleaned and digested with the SP3 method as described previously with modifications^{68,69}. Briefly, 50 µL of a 20 µg/µL SP3 bead stock (Sera-Mag SpeedBead carboxylate-modified magnetic particles; GE Healthcare Life Sciences) and 500 µL acetonitrile (ACN; final concentration of 70%) were added to 150 µL of protein extract and incubated at room temperature for 10 min (1,000 rpm). Tubes were mounted on a magnetic rack, supernatants were removed and beads were washed twice with 70% ethanol and once with ACN (1 mL each). Beads were resuspended in 80 µL 100 mM TEAB and digested for 4 h with LysC followed by tryptic digestion overnight (1:50 protease:protein ratio, 37 °C, 1,000 rpm). Peptides were cleaned by addition of 3.5 µL formic acid (final concentration of 4%) and 1.7 mL ACN (final concentration of 95%) followed by incubation for 10 min. After spinning down (1,000 g) tubes were mounted on a

magnetic rack and beads were washed once with 1.5 mL ACN. Peptides were eluted from the beads with 100 μ L 2% DMSO and acidified with 5.2 μ L 20 % formic acid (final concentration of 1%) followed by centrifugation (15,000 g). Peptide amounts were quantified using the fluorometric CBQCA assay (Thermo Scientific).

TMT labelling

For each sample 15 μ g peptides per sample were dried *in vacuo* in a Concentrator plus (Eppendorf) and resuspended in 50 μ L 200 mM EPPS pH 8.5. TMT10plex tags (Thermo Scientific) were dissolved in anhydrous ACN and added to the peptide sample in a 1:10 peptide:TMT ratio. Additional anhydrous ACN was added to a final volume of 22 μ L. Samples were incubated for 2 h (22 °C, 750 rpm). Unreacted TMT was quenched by incubation with 5 μ L 5% hydroxylamine for 30 min. Samples were combined, dried *in vacuo* and resuspended in 1% TFA followed by clean-up with solid-phase extraction using Waters Sep-Pak tC18 50 mg. Samples were loaded, washed five times with 1 mL 0.1% TFA in water and peptides were eluted with 70% ACN/0.1% TFA (1 mL) and dried *in vacuo* in a Concentrator plus (Eppendorf).

High pH reversed phase peptide fractionation

TMT labelled peptide samples were fractionated using off-line high pH reverse phase chromatography. Dried samples were resuspended in 5% formic acid and loaded onto a 4.6 x 250 mm XBridge BEH130 C18 column (3.5 μ m, 130 Å; Waters). Samples were separated on a Dionex Ultimate 3000 HPLC system with a flow rate of 1 mL/min. Solvents used were water (A), ACN (B) and 100 mM ammonium formate pH 9 (C). While solvent C was kept constant at 10%, solvent B started at 5% for 3 min, increased to 21.5% in 2 min, 48.8% in 11 min and 90% in 1 min, was kept at 90% for further 5 min followed by returning to starting conditions and re-equilibration for 8 min. Peptides were separated into 48 fractions, which were concatenated into 24 fractions and subsequently dried *in vacuo*. Peptides were redissolved in 5% formic acid and analysed by LC-MS.

LC-MS analysis

TMT labelled samples were analysed on an Orbitrap Fusion Tribrid mass spectrometer coupled to a Dionex RSLCnano HPLC (Thermo Scientific). Samples were loaded onto a 100 μ m x 2 cm Acclaim PepMap-C18 trap column (5 μ m, 100 Å) with 0.1% trifluoroacetic acid for 7 min and a constant flow of 4 μ L/min. Peptides were separated on a 75 μ m x 50 cm EASY-Spray C18 column (2 μ m, 100 Å; Thermo Scientific) at 50 °C using a linear gradient from 10% to 40% B in 153 min with a flow rate of 200 nL/min. Solvents used were 0.1% formic acid (A) and 80% ACN/0.1% formic acid (B). The spray was initiated by applying 2.5 kV to the EASY-Spray emitter. The ion transfer capillary temperature was set to 275 °C and the radio frequency of the S-lens to 50%. Data were acquired under the control of Xcalibur software in a data-dependent mode. The number of dependent scans was 12. The full scan was acquired in the orbitrap covering the mass range of m/z 350 to 1,400 with a mass resolution of 120,000, an AGC target of 4×10^5 ions and a maximum injection time of 50 ms. Precursor ions with charges between 2 and 7 and a minimum intensity of 5×10^3 were selected with an isolation window of m/z 1.2 for fragmentation using collision-induced dissociation in the ion trap with 35% collision energy. The ion trap scan rate was set to “rapid”. The AGC target was set to 1×10^4 ions with a maximum injection time of 50 ms and a dynamic exclusion of 60 s. During the MS3 analysis, for more accurate TMT quantification, 5 fragment ions were co-isolated using synchronous precursor selection in a window of m/z 2 and further fragmented with a HCD collision energy of 65%. The fragments were then analysed in the orbitrap with a resolution of 50,000. The AGC target was set to 5×10^4 ions and the maximum injection time was 105 ms.

TMT channel allocation

TMT tag	Cell line
126C	bubh_3 (hiPSC - female)
127N	H01 (hESC - male)
127C	kucg_2 (hiPSC - male)
128N	H09 (hESC - female)
128C	oaqd_3 (hiPSC - male)
129N	SA121 (hESC - male)
129C	wibj_2 (hiPSC - female)
130N	SA181 (hESC - male)
130C	Empty
131	Empty

High-resolution respirometry in wibj_2 and H1 stem cells

Mitochondrial respiration was studied in digitonin-permeabilised wibj_2 and H1 stem cells ($10\ \mu\text{g} / 1 \times 10^6$ cells), to keep mitochondria in their architectural environment. The analysis was performed in an oxygraphic chamber with thermostat set to 37°C with continuous stirring (Oxygraph-2 k, Oroboros instruments, Innsbruck, Austria). Cells were collected with trypsin, pelleted, and then placed in MiR05 respiration medium (110 mM sucrose, 60 mM lactobionic acid, 0.5 mM EGTA, 3 mM MgCl_2 , 20 mM taurine, 10 mM KH_2PO_4 , 20 mM HEPES adjusted to pH 7.1 with KOH at 30°C , and 1 g/l BSA, essentially fatty acid free). Substrate-Uncoupler-Inhibitor titration protocol number 2 (SUIT-002)⁷⁰ was used to determine respiratory rates. Briefly, after residual oxygen consumption in absence of endogenous fuel substrates (ROX, in presence of 2.5 mM ADP), was measured, fatty acid oxidation pathway state (F) was evaluated by adding malate (0.1 mM) and octanoyl carnitine (0.2 mM) (OctM_p). Membrane integrity was tested by adding cytochrome c (10 μM) (OctMc_p). Subsequently, the NADH electron transfer-pathway state (FN) was studied by adding a high concentration of malate (2 mM, OctM_p), pyruvate (5 mM, OctPM_p), and glutamate (10 mM, OctPGM_p). Then succinate (10 mM, OctPGMS_p) was added to stimulate the S pathway (FNS), followed by glycerophosphate (10 mM, OctPGMSG_p) to reach convergent electron flow in the FNSG_p-pathway to the Q-junction. Uncoupled respiration was next measured by performing a titration with CCCP (OctPGMSG_p), followed by inhibition of complex I (SG_p) with rotenone (0.5 μM , SG_p). Finally, residual oxygen consumption (ROX) was measured by adding Antimycin A (2.5 μM). ROX was then subtracted from all respiratory states, to obtain mitochondrial respiration. Results are expressed in $\text{pmol} \cdot \text{s}^{-1} \cdot 1 \times 10^6$ cells. The P/E control ratio, which reflects the control by coupling and limitation by the phosphorylation system, was subsequently calculated by dividing the OctPGMSG_p value by the OctPGMSG_p value.

Radiolabelled glutamine uptake (protocol was adapted from⁷¹)

Two hiPSC lines (wibj_2 and oaqd_3) with 3 technical replicates each were compared to two hESC lines (SA121 and SA181) with 3 technical replicates of each. Both hiPSCs and hESCs were plated in 6-well plates 2 days before the transport assay (5×10^4 cells/cm² – this gives 1×10^6 cells/well on “uptake day”). The cell growth media was carefully aspirated so as not to disturb the adherent monolayer of cells. They were washed gently by pipetting with 5 mls preheated (37°C) uptake solution (HBSS (pH 7.4), GIBCO) and aspirating off. This was repeated 3 times. They were then incubated with 0.5 ml of uptake solution containing [³H]glutamine (5 µCi/ml; Perkin Elmer, NET 55100) in either the presence, or absence, of L-glutamine (5 mM; Sigma) for 2 min.

Glutamine uptake was stopped by removing the uptake solution and washing cells with 2 ml of ice-cold stop solution (HBSS with 10 mM nonradioactive L-glutamine) three times. After the third wash, the cells were lysed in 200 µl of 0.1% SDS and 100 mM NaOH, and 100 µl was used to measure the radioactivity associated with the cells. Finally 100 µl sample was added to scintillation vials containing 3 mls scintillant (*OptiPhase HiSafe 3*, Perkin Elmer). β-radioactivity was measured with Tri-Carb 4910TR liquid scintillation counter.

The net glutamine CPM values were calculated by subtracting the Quench CPM values from the Glutamine CPM values.

TEM Sample Preparation

wibj_2 and H1 cells were fixed on the dish in 4% paraformaldehyde and 2.5% glutaraldehyde in 0.1M sodium cacodylate buffer (pH 7.2) for 30 minutes then scraped and transferred to a tube and fixed for a further 30 minutes prior to pelleting. The pellets were cut into small pieces, washed 3 times in cacodylate buffer and then post-fixed in 1% OsO₄ with 1.5% Na ferricyanide in cacodylate buffer for 60 min. After another 3 washes in cacodylate buffer they were contrasted with 1% tannic acid and 1% uranyl acetate. The cell pellets were then dehydrated through alcohol series into 100% ethanol, changed to propylene oxide left overnight in 50% propylene oxide 50% resin and finally embedded in 100% Durcupan resin (Sigma). The resin was polymerised at 60°C for 48hrs and sectioned on a Leica UCT ultramicrotome. Sections were contrasted with 3% aqueous uranyl acetate and Reynolds lead citrate before imaging on a JEOL 1200EX TEM using a SIS III camera.

Proteomics search parameters

The data were searched and quantified with MaxQuant⁷² (version 1.6.7) against the human SwissProt database from UniProt⁷³ (November 2019). The data were searched with the following parameters: type was set to Reporter ion on MS3 with 10plex TMT, stable modification of carbamidomethyl (C), variable modifications of oxidation (M), acetylation (proteins N terminus) and deamidation (NQ). The missed cleavage threshold was set to 2, and the minimum peptide length was set to 7 amino acids. The false discovery rate was set to 1% for positive identification at the protein and peptide spectrum match (PSM) level.

Unique, shared and razor peptides

Peptides which are exclusive to a single protein group are considered unique peptides. Peptides whose sequences match more than one protein group are called shared peptides. Razor peptides are shared peptides whose intensity gets assigned to a single protein group despite matching multiple protein groups.

Data filtering

All protein groups identified with less than either 2 razor or unique peptides or labelled as ‘Contaminant’, ‘Reverse’ or ‘Only identified by site’ were removed from the analysis.

Peptide normalisation

For supplemental figures 2 & 3 peptide intensities were divided by the sum of the intensity from all histone peptides and were multiplied by 1×10^6 .

Copy number calculations

Protein copy numbers were estimated following the “proteomic ruler” method²⁰, but adapted to work with TMT MS3 data. The summed MS1 intensities were allocated to the different experimental conditions according to their fractional MS3 reporter intensities.

Protein content estimations

The protein content was estimated using the following formula: $CN \times LMW$ and then converting the data from Daltons to picograms, where CN is the protein copy number and MW is the protein molecular weight (in Da).

Differential expression analysis

Fold changes and P-values were calculated in R. For individual proteins the p-values were calculated with the bioconductor package LIMMA⁷⁴ version 3.7. The Q-values provided were generated in R using the “qvalue” package version 2.10.0. P-values for protein families and protein complexes were calculated in R using Welch’s T-test.

hiPSC vs hESC overrepresentation analysis

All overrepresentation analysis were done on WebGestalt. The first analysis selected proteins with a fold change > 2 and a q-value < 0.001. The second analysis selected proteins whose fold change was lower than the median minus one standard deviation (0.195) and a q-value < 0.001. Both analyses used all identified proteins with 2 or more razor and unique peptides as a background and required an FDR lower than 0.05.

Peptide coverage figures

The supplemental figures showing the peptide coverage across H1 and H2 histones (Fig. S2 & S3) were generated with Protter⁷⁵.

Data availability

The raw files and the mzTab outputs were uploaded to PRIDE as a full submission under the identifier PXD014502 and are available at <https://www.ebi.ac.uk/pride/archive/projects/PXD014502>

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Author contributions

A.J.B conceived the study, planned the experiments, analysed and interpreted the data. E.G executed all the proteomic sample preparation, and the mass spectrometry experiments. L.V.S performed the glutamine uptake assay. A.R.P performed the TEM experiments. F.S performed the respiration analysis. H.J. performed the EZQ assay and assisted with data interpretation. L.D cultured the hESC and hiPSCs, performed the pluripotency marker western blot and the cell cycle FACS experiments. C.E and E.K.J.H performed the histone western blots. H.Y., M.P and J.S helped to interpret the data. A.I.L, D.A.C and G.F supervised the project and helped to interpret the data. The paper written be A.J.B and A.I.L and edited by all authors.

Declaration of interests

E.G now works for Boehringer Ingelheim Pharma GmbH & Co. KG. A.I.L, M.P and J.S are board members of Tartan Cell Technologies Ltd. M.P and J.S are board members of Glencoe Software Ltd and AIL is a board member of Platinum Informatics Ltd.

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Editors

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Reviewer #1 (Public review):

Summary:

The authors compared four types of hiPSCs and four types of hESCs at the proteome level to determine their differences. Semiquantitative calculations of protein copy number revealed increased protein content in iPSCs. In particular, the results suggest that mitochondria- and cytoplasm-associated proteins in iPSCs reflect to some extent the state of the original differentiated cells. Basically, it contains responses to almost all comments and adds text mainly to the discussion. No additional experiments were performed in the revision, but I believe that future validation using methods other than proteomics would provide more support for the results.

Pros:

Mitochondrial function was verified by high-resolution respirometry, indicating increased ATP-producing capacity of the phosphorylation system in iPSCs.

Weaknesses:

The proteome data in this study may be the result of a simple examination of differences between the clones, and proteome data should be verified using various methods in the future.

<https://doi.org/10.7554/eLife.92025.2.sa3>

Reviewer #2 (Public review):

Summary:

Pluripotent stem cells are powerful tools for understanding development, differentiation, and disease modeling. The capacity of stem cells to differentiate into various cell types holds great promise for therapeutic applications. However, ethical concerns restrict the use of human embryonic stem cells (hESCs). Consequently, induced human pluripotent stem cells (ihPSCs) offer an attractive alternative for modeling rare diseases, drug screening, and regenerative medicine. A comprehensive understanding of ihPSCs is crucial to establish their similarities and differences compared to hESCs. This work demonstrates systematic differences in the reprogramming of nuclear and non-nuclear proteomes in ihPSCs.

Strengths:

The authors employed quantitative mass spectrometry to compare protein expression differences between independently derived ihPSC and hESC cell lines. Qualitatively, protein expression profiles in ihPSC and hESC were found to be very similar. However, when comparing protein concentration at a cellular level, it became evident that ihPSCs express

higher levels of proteins in the cytoplasm, mitochondria, and plasma membrane, while the expression of nuclear proteins is similar between ihPSCs and hESCs. A higher expression of proteins in ihPSCs was verified by an independent approach, and flow cytometry confirmed that ihPSCs had larger cell size than hESCs. The differences in protein expression were reflected in functional distinctions. For instance, the higher expression of mitochondrial metabolic enzymes, glutamine transporters, and lipid biosynthesis enzymes in ihPSCs was associated with enhanced mitochondrial potential, increased ability to uptake glutamine, and increased ability to form lipid droplets.

Weaknesses:

While this finding is intriguing and interesting, the study falls short of explaining the mechanistic reasons for the observed quantitative proteome differences. It remains unclear whether the increased expression of proteins in ihPSCs is due to enhanced transcription of the genes encoding this group of proteins or due to other reasons, for example, differences in mRNA translation efficiency. Another unresolved question pertains to how the cell type origin influences ihPSC proteomes. For instance, whether ihPSCs derived from fibroblasts, lymphocytes, and other cell types all exhibit differences in their cell size and increased expression of cytoplasmic and mitochondrial proteins. Analyzing ihPSCs derived from different cell types and by different investigators would be necessary to address these questions.

<https://doi.org/10.7554/eLife.92025.2.sa2>

Reviewer #3 (Public review):

This study provides a useful insight into the proteomic analysis of several human induced pluripotent (hiPSC) and human embryonic stem cell (hESC) lines. Although the study is largely descriptive with limited validation of the differences found in the proteomic screen, the findings provide a solid platform for further mechanistic discovery.

<https://doi.org/10.7554/eLife.92025.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors compared four types of hiPSCs and four types of hESCs at the proteome level to elucidate the differences between hiPSCs and hESCs. Semi-quantitative calculations of protein copy numbers revealed increased protein content in iPSCs. Particularly in iPSCs, proteins related to mitochondrial and cytoplasmic were suggested to reflect the state of the original differentiated cells to some extent. However, the most important result of this study is the calculation of the protein copy numbers per cell, and the validity of this result is problematic. In addition, several experiments need to be improved, such as using cells of different genders (iPSC: female, ESC: male) in mitochondrial metabolism experiments.

Strengths:

The focus on the number of copies of proteins is exciting and appreciated if the estimated calculation result is correct and biologically reproducible.

Weaknesses:

The proteome results in this study were likely obtained by simply looking at differences between clones, and the proteome data need to be validated. First, there were only a few clones for comparison, and the gender and number of cells did not match between ESCs and iPSCs. Second, no data show the accuracy of the protein copy number per cell obtained by the proteome data.

We agree with the reviewer that it would be useful to have data from more independent stem cell clones and ideally an equal gender balance of the donors would be preferable. As usual, practical cost-benefit, and time available affect the scope of work that can be performed. We note that the impact of biological donor sex on proteome expression in iPSC lines has already been addressed in previous studies¹³. We will however revise the manuscript to include specific mention of these limitations and propose a larger-scale follow-up when resources are available.

Regarding the estimation of protein copy numbers in our study, we would like to highlight that the proteome ruler approach we have used has been employed extensively in the field previously, with direct validation of differences in copy numbers provided using orthogonal methods to MS, e.g., FACS^{2-4,7,10}. Furthermore, the original manuscript¹⁴ directly compared the copy numbers estimated using the “proteomic ruler” to spike-in protein epitope signature tags and found remarkable concordance. This original study was performed with an older generation mass spectrometer and reduced peptide coverage, compared with the instrumentation used in our present study. Further, we noted that these authors predicted that higher peptide coverage, such as we report in our study, would further increase quantitative performance.

Reviewer #2 (Public Review):

Summary:

Pluripotent stem cells are powerful tools for understanding development, differentiation, and disease modeling. The capacity of stem cells to differentiate into various cell types holds great promise for therapeutic applications. However, ethical concerns restrict the use of human embryonic stem cells (hESCs). Consequently, induced human pluripotent stem cells (ihPSCs) offer an attractive alternative for modeling rare diseases, drug screening, and regenerative medicine. A comprehensive understanding of ihPSCs is crucial to establish their similarities and differences compared to hESCs. This work demonstrates systematic differences in the reprogramming of nuclear and non-nuclear proteomes in ihPSCs.

We thank the reviewer for the positive assessment.

Strengths:

The authors employed quantitative mass spectrometry to compare protein expression differences between independently derived ihPSC and hESC cell lines. Qualitatively, protein expression profiles in ihPSC and hESC were found to be very similar. However, when comparing protein concentration at a cellular level, it became evident that ihPSCs express higher levels of proteins in the cytoplasm, mitochondria, and plasma membrane, while the expression of nuclear proteins is similar between ihPSCs and hESCs. A higher expression of proteins in ihPSCs was verified by an independent approach, and flow

cytometry confirmed that ihPSCs had larger cell sizes than hESCs. The differences in protein expression were reflected in functional distinctions. For instance, the higher expression of mitochondrial metabolic enzymes, glutamine transporters, and lipid biosynthesis enzymes in ihPSCs was associated with enhanced mitochondrial potential, increased ability to uptake glutamine, and increased ability to form lipid droplets.

Weaknesses:

While this finding is intriguing and interesting, the study falls short of explaining the mechanistic reasons for the observed quantitative proteome differences. It remains unclear whether the increased expression of proteins in ihPSCs is due to enhanced transcription of the genes encoding this group of proteins or due to other reasons, for example, differences in mRNA translation efficiency. Another unresolved question pertains to how the cell type origin influences ihPSC proteomes. For instance, whether ihPSCs derived from fibroblasts, lymphocytes, and other cell types all exhibit differences in their cell size and increased expression of cytoplasmic and mitochondrial proteins. Analyzing ihPSCs derived from different cell types and by different investigators would be necessary to address these questions.

We agree with the Reviewer that our study does not extend to also providing a detailed mechanistic explanation for the quantitative differences observed between the two stem cell types and did not claim to have done so. We have now included an expanded section in the discussion where we discuss potential causes. However, in our view fully understanding the reasons for this difference is likely to involve extensive future in-depth analysis in additional studies and is not something that can be determined just by one or two additional supplemental experiments.

We also agree studying hiPSCs reprogrammed from different cell types, such as blood lymphocytes, would be of great interest. Again, while we agree it is a useful way forward, in practice this will require a very substantial additional commitment of time and resources. We have now included a section discussing this opportunity within the discussion to encourage further research into the area.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) *aizi1* and *ueah1* clones, which were analyzed in Figure 1A, were excluded from the proteome analysis. In particular, the *GAPDH* expression level of the *aizi1* clone is similar to that of ESCs and different from other iPSC clones. An explanation of how the clones were selected for proteome analysis is needed. Previously, the comparative analysis of iPSCs and ESCs reported in many studies from 2009-2017 (Ref#1-7) has already shown that the number of clones used in the comparative analysis is small, claiming differences (Ref#1-3) and that the differences become indistinguishable when the number of clones is increased (Ref#4-7). Certainly, few studies have been done at the proteome level, so it is important to examine what differences exist in the proteome. Also, it is interesting to focus on the amount of protein per cell. However, if the authors want to describe biological differences, it would be better to get the proteome data in biological duplicate and state the reason for selecting the clones used.

(1) M. Chin, *Cell Stem Cell*, 2009, PMID: 19570518

(2) K. Kim, *Nat Biotechnol.*, 2011, PMID: 22119740

(3) R. Lister, *Nature*, 2011, PMID: 21289626

(4) A.M. Newman, *Cell Stem Cell*, 2010, PMID: 20682451

(5) M.G. Guenther, *Cell Stem Cell*, 2010, PMID: 20682450

(6) C. Bock, *Cell*, 2010, PMID: 21295703

(7) S. Yamanaka, *Cell Stem Cell*, PMID: 22704507

We agree with the reviewer that analysing more clones would be beneficial. We have included a section of this topic in the discussion. In our study, we only had access to the 4 hESC lines included, therefore in the original proteomic study we also analysed 4 hiPSC lines, which were routinely grown within our stem cell facility. While as the study progressed the stem cell facility expanded the culture of additional hiPSC lines, unfortunately we couldn't also access additional hESC lines.

We agree that ideally combining each biological replicate with additional technical replicates would provide extra robustness. As usual, cost and practical considerations at the time the experiments were performed affected the experimental design chosen. For the experimental design, each experiment was contained within 1 batch to avoid the strong batch effects present in TMT (Brenes et al 2019).

(2) iPSC samples used in the proteome analysis are two types of female and two types of male, while ESC samples are three types of female and one type of female. The number of sexes of the cells in the comparative analysis should be matched because sex differences may bias the results.

While we agree with the reviewer in principle, we have previously performed detailed comparisons of proteome expression in many independent iPSC lines from both biological male and female donors (see Brenes et al., *Cell Reports* 2021) and it seems unlikely that biological sex differences alone could account for the proteome differences between iPS and ESC lines uncovered in this study. However, as this is a relevant point, we have revised the manuscript to explicitly mention this caveat within the discussion section.

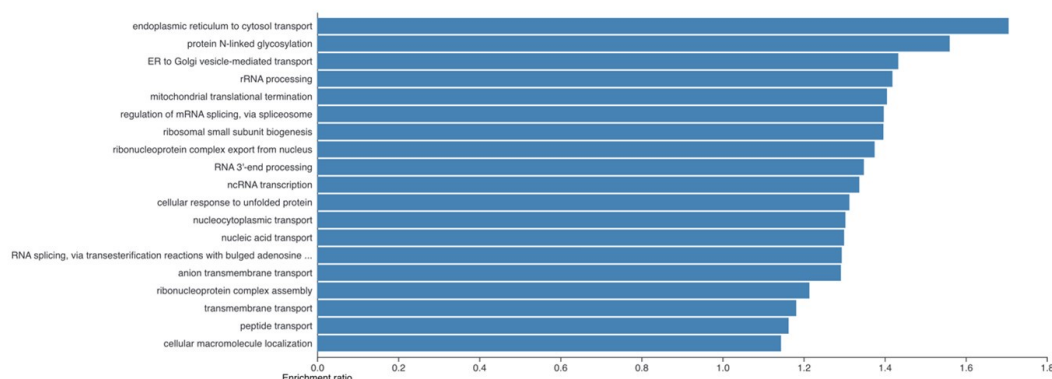
(3) In Figure 1h, I suspect that the variation of PCA plots is very similar between ESCs and iPSCs. In particular, the authors wrote "copy numbers for all 8 replicates" in the legend, but if Figure 1b was done 8 times, there should be 8 types of cells x 8 measurements = 64 points. Even if iPSCs and ESCs are grouped together, there should be 8 points for each cell type. Is it possible that there is only one TMT measurement for this analysis? If so, at least technical duplicates or biological duplicates would be necessary. I also think each cell should be plotted in the PCA analysis instead of combining the four types of ESCs and iPSCs into one.

We thank the reviewer for bringing this error to our attention. The legend has been corrected to state, "for all 8 stem cell lines". Each dot represents the proteome of each of the 4 hESCs and 4 hiPSCs that were analysed using proteomics.

(4) It is necessary to show what functions are enriched in the 4408 proteins whose protein copies per cell were increased in the iPSCs obtained in Figure 2B.

The enrichment analysis requested has been performed and is now included as a new supplemental figure 2. We find it very interesting that despite the large number of proteins involved here (4,408), the enrichment analysis still shows clear enrichment for specific cellular processes. The summary plot using affinity propagation within webgestalt is included here:

Author response image 1.



(5) The Proteomic Ruler method used in this study is a semi-quantitative method to calculate protein copy numbers and is a concentration estimation method. Therefore, if the authors want to have a biological discussion based on the results, they need to show that the estimated concentrations are correct. For example, there are Western Blotting (WB) results for genes with no change in protein levels in hESC and hiPSC in Fig. 6ij, but the WB results for the group of genes that are claimed to have changed are not shown throughout the paper. Also, there is no difference in the total protein level between iPSCs and ESCs from the ponceau staining in Fig.6ij. WB results for at least a few genes are needed to show whether the concentration estimates obtained from the proteome analysis are plausible. If the protein per cell is increased in these iPSC clones, performing WB analysis using an equal number of cells would be better.

Regarding the ‘proteome ruler’ approach we would like to highlight that this method has previously been used extensively in the field, with detailed validation, as already explained above. It is also not ‘semi-quantitative’ and can estimate absolute abundance, as well as concentrations. Our work does not use their concentration formulas, but the estimation of protein copy numbers, which was shown to closely match the observed copy numbers as determined when spike-ins are used¹⁴.

In providing here additional validation using Western Blotting (WB), we prioritised for analysis also by WB the proteins related to pluripotency markers, which are vital to determine the pluripotency state of the hESCs and hiPSCs, as well as histone markers. We have included a section in the discussion concerning additional validation data and agree in general that further validation is always useful.

(6) Regarding the experiment shown in Figure 4l, the gender of iPSC used (wibj2) is female and WA01 (H1; WA01) is male. Certainly, there is a difference in the P/E control ratio, but isn't this just a gender difference? The sexes of the cells need to be matched.

We accept that ideally the sexes of donors should ideally have been matched and have mentioned this within the discussion. Nonetheless, as previously mentioned, our previous detailed proteomic analyses of multiple hiPSC lines¹³ derived from both biological male and female donors provide relevant evidence that the results shown in this study are not simply a reflection of the sex of the donors for the respective iPSC and ESC lines. When comparing eroded and non-eroded female hiPSCs to male hiPSCs we found no significant differences in any electron transport chain proteins, not TCA proteins between males and females.

Minor comments:

(1) Method: Information on the hiPSCs and hESCs used in this study should be described. In particular, the type of differentiated cells, gender, and protocols that were used in the reprogramming are needed.

We agree with the reviewer on this. The hiPSC lines were generated by the HipSci consortium, as described in the flagship HipSci paper¹⁵. We cite the flagship paper, which specifies in great detail the reprogramming protocols and quality control measures, including analysis of copy number variations¹⁵. However, we agree that this information may not be easily accessible for readers. We agree it is relevant to explicitly include this information in our present manuscript, instead of expecting readers to look at the flagship paper. These details have therefore been added to the revised version.

(2) Method: In Figure 1a, Figure 6i, j, the antibody information of Nanog, Oct4, Sox2, and Gapdh is not written in the method and needs to be shown.

The data relating to these has now been included within the methods section.

(3) Method: In Figure 1b and other figures, the authors should indicate which iPSC corresponds to which TMT label; the data in the Supplemental Table also needs to indicate which data is which clone.

We have now added this to the methods section.

(4) Method: The method of the FACS experiment used in Figure 2 should be described.

The methods related to the FACS analysis have now been included within the manuscript.

(5) Method: The cell name used in the mitochondria experiment shown in Figure 4 is listed as WA01, which is thought to be H1. Variations in notation should be corrected.

This has now been corrected.

(6) Method: The name of the cell clone shown in Figure 3l,m should be mentioned.

We have now added these details on the corresponding figure and legend.

Reviewer #2 (Recommendations For The Authors):

This study utilized quantitative mass spectrometry to compare protein expression in independently derived 4 hiPSC and 4 hESC cell lines. The investigation quantified approximately 7,900 proteins, and employing the "Proteome ruler" approach, estimated protein copy numbers per cell. Principal component analyses, based on protein copy number per cell, clearly separated hiPSC and hESC, while different hiPSCs and hESCs grouped together. The study revealed a global increase in the expression of cytoplasmic, mitochondrial, membrane transporters, and secreted proteins in hiPSCs compared to hESCs. Interestingly, standard median-based normalization approaches failed to capture these differences, and the disparities became apparent only when protein copy numbers were adjusted for cell numbers. Increased protein abundance in hiPSC was associated with augmented ribosome biogenesis. Total protein content was >50% higher in hiPSCs compared to hESCs, a observation independently verified by total protein content measurement via the EZQ assay and further supported by the larger cell size of hiPSCs in flow cytometry. However, the cell cycle distribution of hiPSC and hESC was similar,

indicating that the difference in protein content was not due to variations in the cell cycle. At the phenotypic level, differences in protein expression also correlated with increased glutamine uptake, enhanced mitochondrial potential, and lipid droplet formation in hiPSCs. iPSCs also expressed higher levels of extracellular matrix components and growth factors.

Overall, the presented conclusions are adequately supported by the data. Although the mechanistic basis of proteome differences in iPSC and hESC is not investigated, the work presents interesting findings that are worthy of publication. Below, I have listed my specific questions and comments for the authors.

(1) Figure 1a displays immunoblots from 6 iPSC and 4 ESC cell lines, with 8 cell lines (4 hESC, 4 hiPSC) utilized in proteomic analyses (Fig. 1b). The figure legend should specify the 8 cell lines included in the proteomic analyses. The manuscript text describing these results should explicitly mention the number and names of cell lines used in these assays.

We agree with the reviewer and have now marked in figure 1 all the lines that were used for proteomics and have added a section in the methods specifying which cell lines were analysed in each TMT channel.

(2) In most figures, the quantitative differences in protein expression between hiPSC and hESC are evident, and protein expression is highly consistent among different hiPSCs and hESCs. However, the glutamine uptake capacity of different hiPSC cell lines, and to some extent hESC cell lines, appears highly variable (Figure 3e). While proteome changes were measured in 4 hiPSCs and 4 hESCs, the glutamine uptake assays were performed on a larger number of cell lines. The authors should clarify the number of cell lines used in the glutamine uptake assay, clearly indicating the cell lines used in the proteome measurements. Given the large variation in glutamine uptake among different cell lines, it would be useful to plot the correlation between the expression of glutamine transporters and glutamine uptake in individual cell lines. This may help understand whether differences in glutamine uptake are related to variations in the expression of glutamine transporters.

The “proteomic ruler” has the capacity to estimate the protein copy numbers per cell, as such changes in the absolute number of cells that were analysed do not cause major complications in quantification. Furthermore, TMT-based proteomics is the most precise proteomics methods available, where the same peptides are detected in all samples across the same data points and peaks, as long as the analysis is done within a single batch, as is the case here.

The glutamine uptake assay is much more sensitive to the variation in the number of cells. The number of cells were estimated by plating the cells with approximately 5e4 cells two days before the assay, which creates variability. Furthermore, hESCs and hiPSCs are more adhesive than the cells used in the original protocol, hence the quench data was noisier for these lines, making the data from the assay more variable.

(3) In Figure 4j, it would be helpful to indicate whether the observed differences in the respiration parameters are statistically significant.

We have now modified the plot to show which proteins were significantly different.

(4) The iPSCs used here are generated from human primary skin fibroblasts. Different cells vary in size; for instance, fibroblast cells are generally larger than blood lymphocytes. This raises the question of whether the parent cell origin impacts differences in hiPSCs and hESC proteomes. For example, do the authors anticipate that hiPSCs derived from small somatic cells would also display higher expression of

cytoplasmic, mitochondrial, and membrane transporters compared to ESC? The authors may consider discussing this point.

This is a very interesting point. We have now added an extension to the discussion focussed on this subject.

(5) One wonders if the "Proteome ruler" approach could be applied retrospectively to previously published ihPSC and hESC proteome data, confirming higher expression of cytoplasmic and mitochondrial proteins in ihPSCs, which may have been masked in previous analyses due to median-based normalization.

We agree with the reviewer and think this is a very good suggestion. Unfortunately, in the main proteomic papers comparing hESC and hiPSCs^{16,17} the authors did not upload their raw files to a public repository (as it was not mandatory at that period in time), and they also used the International Protein Index (IPI), which is a discontinued database. So the raw files can't be reprocessed and the database doesn't match the modern SwissProt entries. Therefore, reprocessing the previous data was impractical.

(6) The work raises a fundamental question: what is the mechanistic basis for the higher expression of cytoplasmic and mitochondrial proteins in ihPSCs? Conceivably, this could be due to two reasons: (a) Genes encoding cytoplasmic and mitochondrial proteins are expressed at a higher level in ihPSCs compared to hESC. (b) mRNAs encoding cytoplasmic and mitochondrial proteins are translated at a higher level in ihPSCs compared to hESC. The authors may check published transcriptome data from the same cell lines to shed light on this point.

This is a very interesting point. We believe that the reprogrammed cells contained mature mitochondria, which are not fully regressed upon reprogramming and that this can establish a growth advantage in the normoxic environments in which the cells are grown. Unfortunately, the available transcriptomic data lacked spike-ins, and thus only enables comparison of concentration, not of copy numbers¹³. Therefore, we could not determine with the available data if there was an increase in the copies of specific mRNAs. However, with a future study where there was a transcriptomic dataset with spike-ins included, this would be very interesting to analyse.

Reviewer #3 (Recommendations For The Authors):

It is unclear whether changes in protein levels relate to any phenotypic features of cell lines used. For example, the authors highlight that increased protein expression in hiPSC lines is consistent with the requirement to sustain high growth rates, but there is no data to demonstrate whether hiPSC lines used indeed have higher growth rates.

We respectfully disagree with the reviewer on this point. Our data show that hESCs and hiPSCs show significant differences in protein mass and cell size, with the MS data validated by the EZQ assay and FACS, while having no significant differences in their cell cycle profiles. Thus, increased size and protein content would require higher growth rates to sustain the increased mass, which is what we observe.

The authors claim that the cell cycle of the lines is unchanged. However, no details of the method for assessing the cell cycle were included so it is difficult to appreciate if this assessment was appropriately carried out and controlled for.

We apologise for this omission; the details have been included in the revised version of the manuscript.

Details and characterisation of iPSC and ESC lines used in this study are overall lacking. The lines used are merely listed in methods, but no references are included for published lines, how lines were obtained, what passage they were used at, their karyotype status etc. For details of basic characterisation, the authors should refer to the ISSC Standards for the use of human stem cells in research. In particular, the authors should consider whether any of the changes they see may be attributed to copy number variants in different lines.

We agree with the reviewer on this and refer to the reply above concerning this issue.

The expression data for markers of undifferentiated state in Figure 1a would ideally be shown by immunocytochemistry or flow cytometry as it is impossible to tell whether cultures are heterogeneous for marker expression.

We agree with the reviewer on this. FACS is indeed much more quantitative and a better method to study heterogeneity. However, we did not have protocols to study these markers using FACS.

TEM analysis should ideally be quantified.

We agree with the reviewer that it would be nice to have a quantitative measure.

All figure legends should explicitly state what graphs are representing (e.g. average/mean; how many replicates (biological or technical), which lines)? Some data is included in Methods (e.g. glutamine uptake), but not for all of the data (e.g. TEM).

We agree with the reviewer. These has been corrected in the revised version of the manuscript, with additional details included.

Validation experiments were performed typically on one or two cell lines, but the lines used were not consistent (e.g. wibj_2 versus H1 for respirometry and wibj_2, oaqd_3 versus SA121 and SA181 for glutamine uptake). Can the authors explain how the lines were chosen?

The validation experiments were performed at different time points, and the selection of lines reflected the availability of hiPSC and hESC lines within our stem cell facility at a given point in time.

We chose to use a range of different lines for comparison, rather than always comparing only one set of lines, to try to avoid a possible bias in our conclusions and thus to make the results more general.

The authors should acknowledge the need for further functional validation of the results related to immunosuppressive proteins.

We agree with the reviewer and have added a sentence in the discussion making this point explicitly.

Differences in H1 histones abundance were highlighted. Can the authors speculate as to the meaning of these differences?

Regarding H1 histones, our study of the literature, as well as discussions with with chromatin and histone experts, both within our institute and externally, have not shed light into what

the differences could imply, based upon previous literature. We think therefore that this is a striking and interesting result that merits further study, but we have not yet been able to formulate a clear hypothesis on the consequences.

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